Lab 08 Answers Report

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Contents

1 Question 1

Define a Prolog knowledge base for the Unix directory structure. Refer to the diagram on the previous slide. Use the predicate location(Item, Folder) to represent that an Item (which can be a file or another folder) is located directly inside a Folder. For the top-level directory /, refer to it as root.

1.1 Implementation

```prolog
% Directory structure
location(bin, root).
location(etc, root).
location(sbin, root).
location(usr, root).
location(var, root).
location(home, root).

% /bin dir
location(bash, bin).
location(cat, bin).
location(ls, bin).

% /etc dir
location(crontab, etc).
location(cups, etc).
location(fonts, etc).
location(fstab, etc).
location('host.conf', etc).

% /usr dir
location(bin, usr).
location(include, usr).
location(lib, usr).
location(local, usr).

% /usr/local dir
location(bin, local).
location(lib, local).
```

Listing 1: UNIX Directory Knowledge Base

2 Question 2

Query the knowledge base to find the parent folder (the direct location) of the fstab file. Ensure that you show the exact Prolog query and the results received.

Result:

Figure 1: Result for `location(fstab, Parent)` query.

3 Question 3

Query the knowledge base to find all of the items directly inside the /etc folder. Ensure that you show the exact Prolog query and the results received

Result:

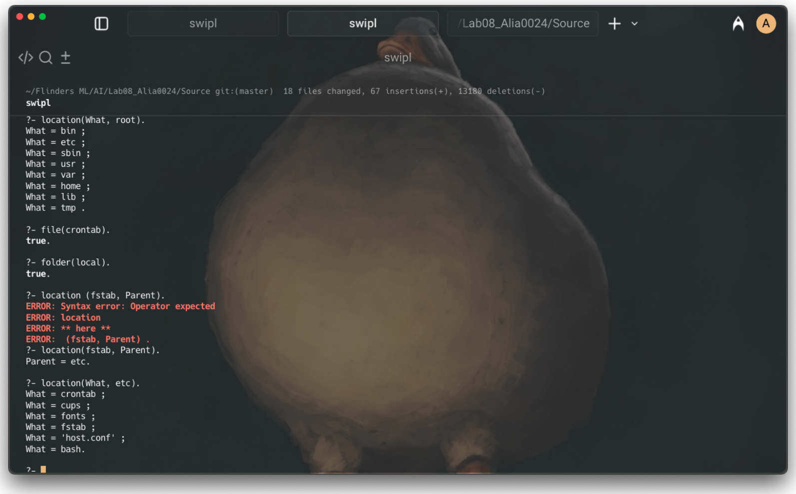

Figure 2: Result for `location(Item, etc)` query.

4 Question 4

First, define a new Prolog rule is_in_grandparent (Item, GrandparentFolder) that is true if an Item is located two levels deep inside the GrandparentFolder. Next, use this rule to find the grandparent folder of /usr/local/bin. Ensure to show the complete rule, the query, and the results.

4.1 Implementation

```

is_in_grandparent(Item, GrandparentFolder) :-
    location(Item, Parent),
    location(Parent, GrandparentFolder).

find_path(Item, Path) :-
    location(Item, Parent),
    Parent \= root,
    find_path(Parent, ParentPath),
    append(ParentPath, [Parent], Path).
```

Result:

Figure 3: Result for `is_in_grandparent(bin, Grandparent)` query.

5 Question 5

Define a recursive Prolog rule find_path(Item, Path) that constructs the full, absolute path for any given Item as a list. The path should start with root. For example, querying for crontab should return Path = [root, etc]. Use your rule to find the path for the /var/log folder. Ensure to show the complete rule(s), the query, and the results.

5.1 Implementation

```

find_path(Item, [root]) :-           % Base case: path is [root]
    location(Item, root).

find_path(Item, Path) :-
    location(Item, Parent),
    Parent \= root,
    find_path(Parent, ParentPath),
    append(ParentPath, [Parent], Path).  % Append [Parent] to build path
```

Result:

Figure 4: Result for `find_path(log, Path)` query.